

Fault-Tolerant Swarm Quadrotor Control for Non-Convex Geodetic Mapping

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Abstract. Surveying a non-convex site with a quadrotor swarm must remain safe and complete when the wind is strong, obstacles are unknown in advance, and agents fail in flight. We present a three-layer architecture for this task. The high level partitions the region by polygon-clipped centroidal Voronoi tessellation, plans boundary-following boustrophedon sweeps and reallocates the unswept lanes of failed agents. The mid level maps obstacles with on-board LiDAR and filters every velocity command through a control barrier function quadratic program whose class- $\mathcal{K}$ function follows from the tilt limit and dead time of the vehicle. The low level is a cascaded PID with gains synthesised from the LQR or the H_∞ Riccati equation. Seven quadrotors were flown in ROS 2/Gazebo over one convex and three non-convex regions under Gauss–Markov turbulence ($\sigma = 2.5$ m/s) over a 3.0 m/s mean wind, nine unmapped static and two moving obstacles, and two scheduled agent losses. Seven of eight missions finished with 96.9–99.0% coverage despite losing 28.6% of the fleet. PID- H_∞ completed all four regions, lowered the RMS lateral sweep error by 16–39%, mostly by rejecting the mean wind, and finished 11–15% sooner than PID-LQR where both finished, at 3.9–5.6 times its torque effort. One PID-LQR mission was lost when an agent touched an obstacle, and both controllers grazed the same obstacle in one region. Each configuration was flown once, so these are single-run observations.

INTRODUCTION

Multi-rotor swarms shorten aerial surveys by flying several sensors in parallel, but real survey sites are rarely rectangles: they have re-entrant corners and narrow arms, gusty wind, obstacles unknown before take-off, and vehicles that fail. Centroidal Voronoi tessellation (CVT) with Lloyd’s iteration [1] distributes agents over a region and has been combined with obstacle avoidance and actuator faults [2], safety constraints [3], non-convex domains [4] and agent dropout [5]; boustrophedon sweeps remain a standard way to cover a cell [6, 7], including cooperative UAV surveys [8] with in-flight re-planning after failures [9]. Control barrier functions (CBF) enforce safety as a quadratic program (QP) on the control input [10, 11], with high-order [12], multi-robot [13] and quadrotor forms for moving obstacles [14] and wind gusts [15]; LQR and H_∞ synthesis remain common routes to multirotor control under wind [16, 17], and fault tolerance in swarms is an active topic [18]. Hierarchical quadcopter swarms with a reactive potential-field planner above cascaded PID loops can fly formations through clutter and wind gusts, yet purely reactive planners remain prone to U-shaped traps [19].

What is less reported is how these pieces behave *together* on a physically simulated vehicle, where a velocity command cannot be realised instantly, a barrier that assumes it can is violated, and a planner that assumes perfect tracking loses coverage to wind drift. This paper contributes (i) an integrated architecture that runs polygon-aware CVT partitioning, boundary-following boustrophedon sweeps and lane-level fault reallocation above a LiDAR-mapped CBF-QP safety filter whose class- $\mathcal{K}$ function is derived from the closed-loop model of the vehicle; (ii) a side-by-side comparison of PID gains synthesised from the LQR and the H_∞ Riccati equations, both on the linear model and inside the full swarm; and (iii) a data-traceable evaluation of eight full missions under combined turbulence, unmapped obstacles, moving obstacles and agent loss, including an analysis of the failures that occurred. Every reported result is computed by an analysis script from the recorded telemetry.

METHODOLOGY

System Architecture

Figure 1 shows the three layers. They are defined per agent: each agent owns a cell, a lane queue, a safety filter that uses neighbours within 6 m and obstacles within 10 m, and a flight controller. Here the high and mid levels of all agents are hosted in one ROS 2 node [20] at 20 Hz, sharing the coverage grid, obstacle map and failure notices without delay, while each vehicle runs its own low level at 250 Hz; distributing the upper layers is future work.

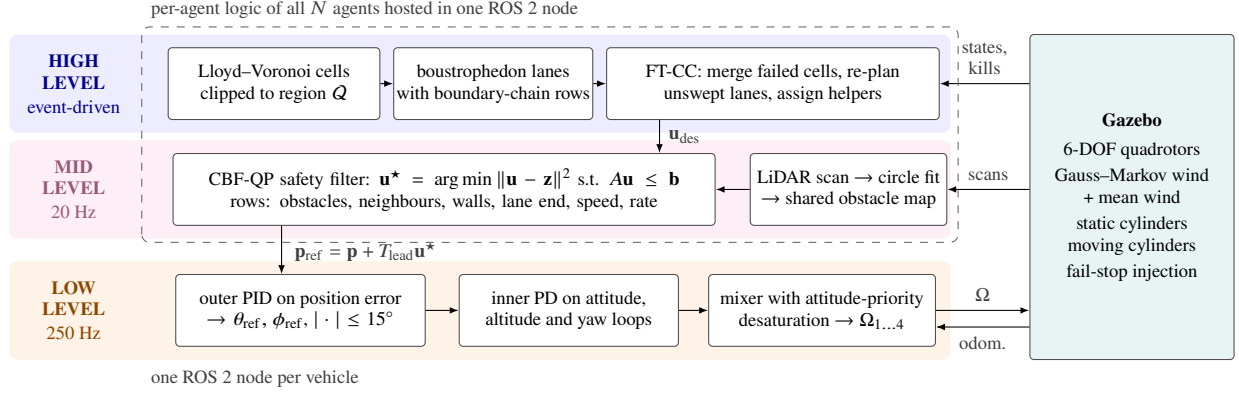


FIGURE 1. Three-layer architecture. The layers are defined per agent; the high and mid levels of all agents run in one ROS 2 node and each vehicle runs its own low level. Plant, wind, obstacles, sensors and faults are simulated in Gazebo.

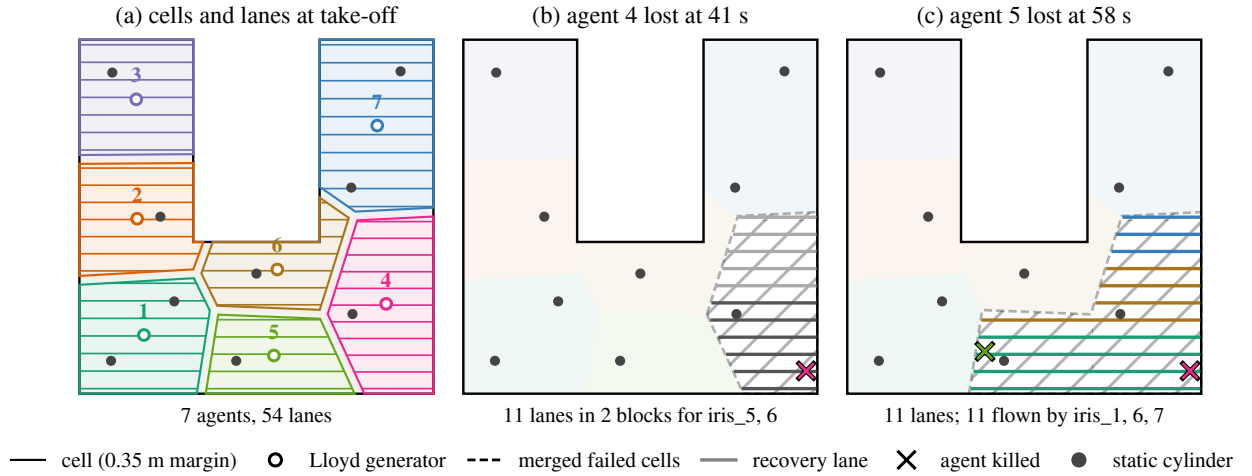


FIGURE 2. High-level planning in the PID- H_∞ U-shape mission, reproduced with the planner code (centroids within 0.6 cm of the log, identical lane counts). (a) Cells after the 0.35 m margin with 54 lanes; each cell's first and last lane follow its boundary. (b) Agent 4 is lost; its cell is re-planned into 11 lanes in blocks for agents 5, 6. (c) Helper 5 is lost too; both cells are merged into 11 lanes for agents 6, 1, 7, coloured by the agent that flew them (from telemetry). Lanes ignore obstacles by design.

High Level: Partition, Sweep and Fault-Tolerant Cooperative Coverage

Given the region polygon Q and generators $\mathbf{g}_i$, the cell of agent i is

$$\mathcal{P}_i = \{\mathbf{q} \in Q : \|\mathbf{q} - \mathbf{g}_i\| \leq \|\mathbf{q} - \mathbf{g}_j\|, \forall j \neq i\}, \quad (1)$$

computed by clipping bisector half-planes and intersecting the result with Q . Lloyd's iteration $\mathbf{g}_i \leftarrow \text{centroid}(\mathcal{P}_i)$ (25 iterations from deterministic k-means++ seeds) minimises $\sum_i \int_{\mathcal{P}_i} \|\mathbf{q} - \mathbf{g}_i\|^2 d\mathbf{q}$ [1], and agents are assigned to cells by the Hungarian algorithm. Each cell is shrunk by 0.35 m and swept by east-west lanes at most 1.45 m apart; its first and last lanes follow the cell boundary, so the wedges that a fixed-pitch sweep leaves at slanted Voronoi edges are flown too (Fig. 2a). With a sensing radius $r_s = 0.95$ m the swath is 1.90 m (overlap of at least 0.45 m). Coverage is scored on a 0.30 m grid: a grid cell inside Q counts once its centre has been within r_s of a live, sweeping agent flying above 0.8 m; grid cells under static obstacles are excluded.

When agent k fails, Fault-Tolerant Cooperative Coverage (FT-CC) keeps the recovery lanes that k still carried, erases the coverage recorded inside $\mathcal{P}_k$ (the data of the lost vehicle are treated as lost), merges all failed cells, re-plans lanes over their union and drops lanes that are already at least 80% covered. The helpers are the neighbours of the failed cells closest to the recovery centroid (one for a single loss with at most five lanes, otherwise two, and three after a second loss). They receive contiguous serpentine blocks, which they fly at once if their own cell is finished and append to their queue otherwise (Fig. 2b,c).

Mid Level: Obstacle Mapping and CBF-QP Safety Filter

Obstacle map. Each 360-ray scan is cleaned of returns within 0.55 m of other agents and 0.80 m of the arena boundary, clustered, and each cluster is fitted with a circle. Detections are associated into one swarm-wide track list; a track is confirmed after 12 hits and rejected if it moves. Static obstacle rows use only confirmed tracks, so the swarm starts with an empty map, and the planner deliberately ignores obstacles, leaving avoidance entirely to the safety filter. The two moving obstacles are supplied from simulator odometry, i.e. they are assumed to be tracked perfectly.

Actuator-consistent barrier. With the reference rule $\mathbf{p}_{\text{ref}} = \mathbf{p} + T_{\text{lead}}\mathbf{u}$, the outer PID makes the commanded-velocity response first order with an acceleration limit, $\dot{v} = \text{sat}_{\pm a_{\text{max}}}(k_v(u - v))$, where

$$k_v = gK_d, \quad a_{\text{max}} = g \tan \theta_{\text{max}}, \quad T_{\text{lead}} = \frac{K_d - k_{\text{ff}}}{K_p}, \quad v_c = \frac{a_{\text{max}}}{k_v}, \quad (2)$$

with $\theta_{\text{max}} = 15^\circ$ and $k_{\text{ff}} = 0.15 \text{ rad s/m}$. With the LQR outer-loop gains (used for both controllers) $k_v = 3.867 \text{ s}^{-1}$, $a_{\text{max}} = 2.631 \text{ m/s}^2$, $T_{\text{lead}} = 0.323 \text{ s}$ and $v_c = 0.680 \text{ m/s}$. The stopping distance from speed $s > v_c$ is $D(s) = (s^2 + v_c^2)/2a$; requiring that the distance travelled during a lumped dead time $T_d = 0.40 \text{ s}$ plus $D(s)$ fit inside the clearance h gives the admissible approach speed

$$\varphi(h) = \begin{cases} \max(0, -aT_d + \sqrt{\max(0, a^2T_d^2 + 2ah - v_c^2)}), & h \geq 0, \\ \gamma_r h, & h < 0, \end{cases} \quad (3)$$

which is exactly zero for $h \leq v_c^2/2a$, unlike a linear γh (Fig. 3a); the branch $h < 0$ ($\gamma_r = 3.5 \text{ s}^{-1}$) demands an outward velocity once the set has been left. The budget is $a = \max(0.15a_{\text{max}}, \mu a_{\text{max}} - |a_{\text{obs}}| - |a_{\text{wind}}|)$ with $\mu = 0.75$ (0.85 for walls) and a wind allowance of 0.5 m/s^2 .

Quadratic program. For an obstacle or neighbour at $\mathbf{o}$ with $\hat{\mathbf{n}} = (\mathbf{p} - \mathbf{o})/\|\mathbf{p} - \mathbf{o}\|$, the condition $\dot{h} \geq -\varphi(h)$ is the linear row

$$-\hat{\mathbf{n}}^\top \mathbf{u} \leq \lambda \varphi(h) - \hat{\mathbf{n}}^\top \mathbf{v}_o, \quad h = \|\mathbf{p} - \mathbf{o}\| - (\rho + r_d + \delta), \quad (4)$$

with drone radius $r_d = 0.22 \text{ m}$, obstacle radius ρ (fitted radius $+0.05 \text{ m}$ for mapped cylinders, 0.50 m for moving ones), margins $\delta = 0.35 \text{ m}$ (static) and 0.75 m (moving), and $\lambda = 1$ for obstacles. Neighbour rows are reciprocal ($\lambda_{ij} + \lambda_{ji} = 1$, weighted by mission state) at a hard radius of 0.70 m and a soft radius of 1.20 m ; walls and the end of the current lane give half-plane rows, and inscribed polygons bound the speed (3 m/s) and the per-tick change of $\mathbf{u}$. Each agent then solves

$$\mathbf{u}^* = \arg \min_{\mathbf{u} \in \mathbb{R}^2} \|\mathbf{u} - \mathbf{z}\|^2 \text{ s.t. } A\mathbf{u} \leq \mathbf{b}, \quad \mathbf{z} = \frac{\mathbf{u}_{\text{des}} + \mathbf{t} + w \mathbf{u}_{\text{prev}}}{1 + w}, \quad (5)$$

exactly, by enumerating the unconstrained point, the projections onto every row and the pairwise vertices (Fig. 3b). A tangential bias $\mathbf{t}$, applied within a 60° cone of a head-on approach, breaks symmetric deadlocks and enters only the objective; $w = 0.25$ smooths successive commands. If the QP is infeasible, soft rows are released (tier 1), then the violation of the hard rows is minimised (tier 2), and a scaled previous command is the last resort (tier 3).

Low Level: LQR and H_∞ Gain Synthesis

Each axis is modelled about hover as a double integrator $\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}$ (e.g. $B = [0, g]^\top$ from pitch to x , $B = [0, 1/I_y]^\top$ from torque to pitch) and augmented with an integrator on the input, $\mathbf{x}_a = [\mathbf{x}^\top, u]^\top$. The LQR design solves the algebraic Riccati equation (ARE) for $K_a = R^{-1}B_a^\top P$. The H_∞ design treats a disturbance entering through the input channel, $B_w = [B^\top, 0]^\top$, and solves

$$A_a^\top P + PA_a + Q_a - P(B_a R^{-1} B_a^\top - \gamma^{-2} B_w B_w^\top)P = 0, \quad (6)$$

rejecting solutions for which $A_a - B_a K_a$ is not Hurwitz [16], with $\gamma = 1.3 \gamma_{\text{min}}$ and γ_{min} found by bisection. The state-feedback gain is mapped to PID form through the pseudo-inverse of $\Gamma = [C^\top, (CA)^\top, (CA^2)^\top; 0, (CB)^\top, (CAB)^\top]$; the synthesised inner integral gain is not used. Figure 4 shows where the gains enter the signal path of one control step, and Table 1 lists weights, attenuation levels and gains.

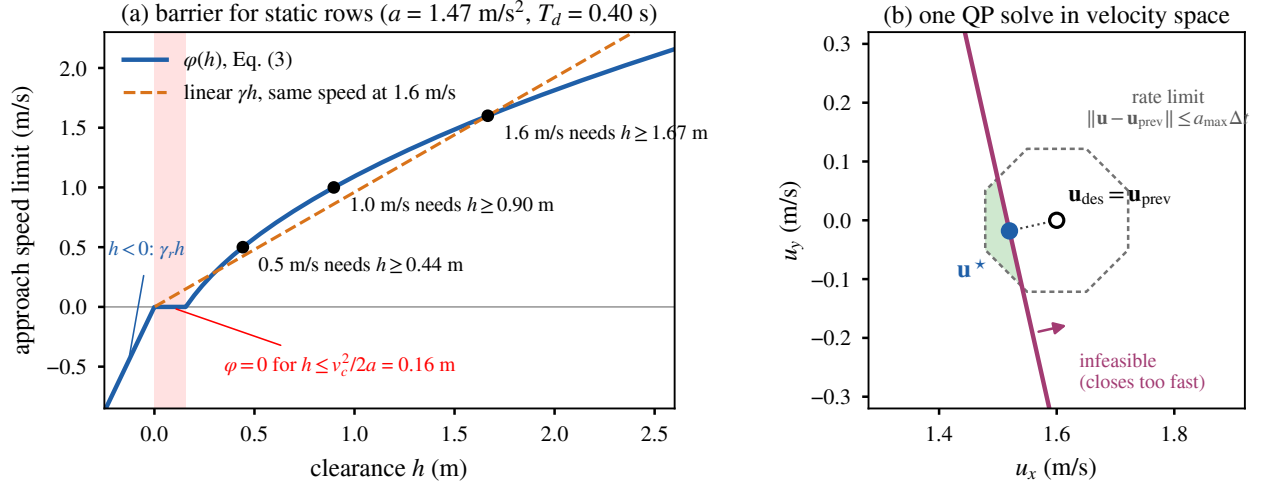


FIGURE 3. Mid-level safety filter with the mission parameters. (a) $\varphi(h)$ of Eq. (3) for static rows: zero below $h = 0.16 \text{ m}$, and $h \geq 1.67 \text{ m}$ is needed to approach at 1.6 m/s ; a linear γh equal at 1.6 m/s permits approach where φ demands a stop. (b) One QP solve by the flight code, illustrative case (east sweep at 1.6 m/s , mapped cylinder 2.51 m ahead, $h = 1.49 \text{ m}$): the row of Eq. (4) cuts the rate-limited set, and the closing speed drops from 1.56 to 1.48 m/s .

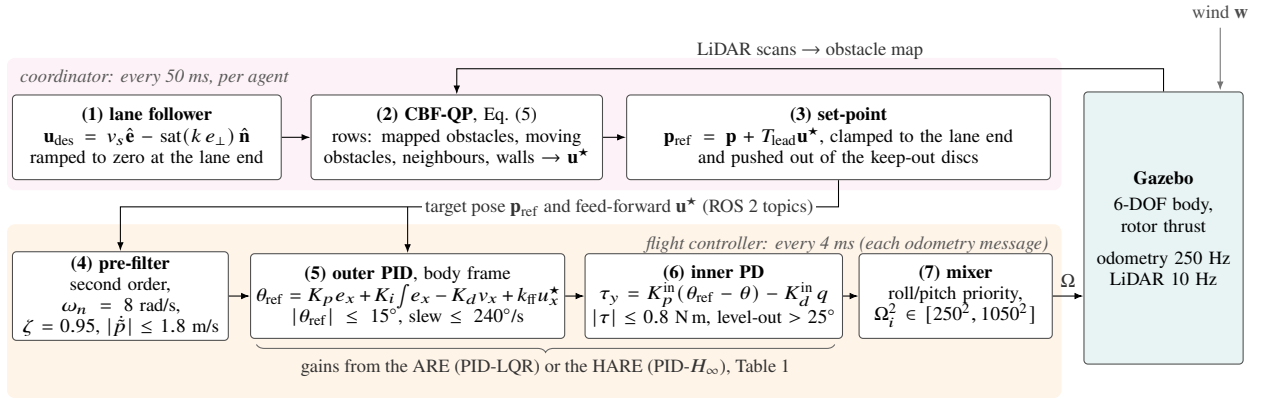


FIGURE 4. Signal path of one control step. The coordinator (1)–(3) publishes a set-point and a velocity feed-forward every 50 ms; the flight controller (4)–(7) runs on each 4 ms odometry message. The controllers differ only in the gains of (5)–(6) and in the position integrator (PID- H_∞ : always active, clipped to ± 0.6 ; PID-LQR: active within 0.25 m, clipped to ± 0.4).

Simulation Setup

Seven 3DR Iris quadrotors ($m = 1.50 \text{ kg}$, $I_{xx} = I_{yy} = 0.0291$, $I_{zz} = 0.0552 \text{ kg m}^2$, arm 0.22 m , $k_f = 8.549 \times 10^{-6} \text{ N s}^2$, rotor speed $\leq 1100 \text{ rad/s}$) are simulated in Gazebo [21] with ROS 2. Controllers read ground-truth odometry; no state estimator is in the loop. Missions start from a launch pad south of the region, cruise at 2.0 m and sweep at 1.6 m/s . Four regions inside a $28 \times 28 \text{ m}$ box are used (Fig. 5): *rect* (784 m^2), *L-shape* (588 m^2), *U-shape* (624 m^2 , 10 m wide notch) and *plus* (460 m^2 , 10 m wide arms), each with nine vertical cylinders ($r = 0.40 \text{ m}$, 4 m tall) placed inside the region. All missions use the same combined-hazard scheme:

- Wind.** A first-order Gauss–Markov process per horizontal axis ($\sigma = 2.5 \text{ m/s}$, correlation time 0.5 s, seed 42), a surrogate for the Dryden and von Kármán models used in quadrotor studies [22], plus a constant -2 m/s mean on both axes from $t = 5 \text{ s}$, uniform in space and applied through the Gazebo wind-effects plugin (vertical component scaled by 0.4). Recorded mean $(-2.34, -1.95) \text{ m/s}$, standard deviation $(2.45, 2.51) \text{ m/s}$, horizontal peaks 12.0 m/s .
- Moving obstacles.** Two kinematic cylinders ($r = 0.45 \text{ m}$) oscillate along the region diagonals with amplitude 10 m (12 m, offset by 1 m, in *plus*) at 0.15 and 0.11 rad/s, i.e. peak speeds of 1.6–2.5 m/s.
- Agent loss.** When coverage first reaches 18% and 32%, one pre-selected agent is killed (rotor speeds set to zero, fail-stop) and the coordinator is notified at once. Victims were chosen to own comparable cells next to the launch pad (Table 2).

TABLE 1. Gain synthesis (outer x/y and inner roll/pitch loops) and linear analysis of the outer loop with the inner loop closed.
 $\|G\|_\infty$: peak gain from a disturbance acceleration to position; PM/GM: phase and gain margins of the outer loop.

Design	Loop	Q, R	$\gamma_{\min} \rightarrow \gamma$	K_p	K_i	K_d	$\ G\ _\infty$ (s ²)	PM	GM
PID-LQR	outer	diag(0.08, 0.35), 1	—	0.757	0.283	0.394	0.158	37.4°	11.5 dB
	inner	diag(6, 0.6), 0.05	—	4.868	—	0.534			
PID- H_∞	outer	diag(0.10, 0.40), 1	2.54→3.30	2.705	1.092	1.107	0.040	25.2°	9.0 dB
	inner	diag(4, 0.4), 0.06	2.54→3.31	11.604	—	1.189			

TABLE 2. Mission results. T : mission time; lat.: RMS lateral sweep error; $\bar{v}$: mean sweep speed; alt.: RMS altitude error; tilt: 95th percentile while sweeping, and maximum; effort: mean $\tau_x^2 + \tau_y^2$ (N²m²); d_{obs} , d_{v2v} : minimum centre-to-surface distance to a static cylinder and minimum inter-agent distance; tier: QP solves needing tier 1 or 2. Killed agents: rect 4, 7 (H_∞), 7 (LQR); L-shape 4, 7; U-shape 4, 5; plus 1, 7. PID-LQR rect covers the 78.8 s before its abort.

Region	Controller	Status	Cov. (%)	T (s)	lat. (cm)	$\bar{v}$ (m/s)	alt. (cm)	tilt ₉₅ (°)	tilt _{max} (°)	effort	d_{obs} (m)	d_{v2v} (m)	tier (%)
rect	PID- H_∞	done	98.2	260	7.2	1.20	2.5	14.7	21.1	0.054	0.49	1.71	2.9
	PID-LQR	abort@78.8	n/a	n/a	11.7	1.12	7.3	14.7	155.9	0.024	0.06	1.52	n/a
L-shape	PID- H_∞	done	98.4	204	8.4	1.19	2.5	14.5	20.8	0.058	0.57	1.32	3.8
	PID-LQR	done	98.5	236	12.2	1.11	2.6	14.5	26.3	0.010	0.29	0.79	4.2
U-shape	PID- H_∞	done	98.5	205	8.3	1.21	2.6	14.8	20.7	0.044	0.63	1.77	4.1
	PID-LQR	done	96.9	240	11.4	1.07	2.5	14.2	16.6	0.011	0.42	1.73	4.1
plus	PID- H_∞	done	98.4	198	9.7	1.15	5.4	14.7	82.5	0.053	0.24	1.24	4.1
	PID-LQR	done	99.0	222	11.5	1.02	2.6	14.4	40.1	0.014	0.24	1.12	4.9

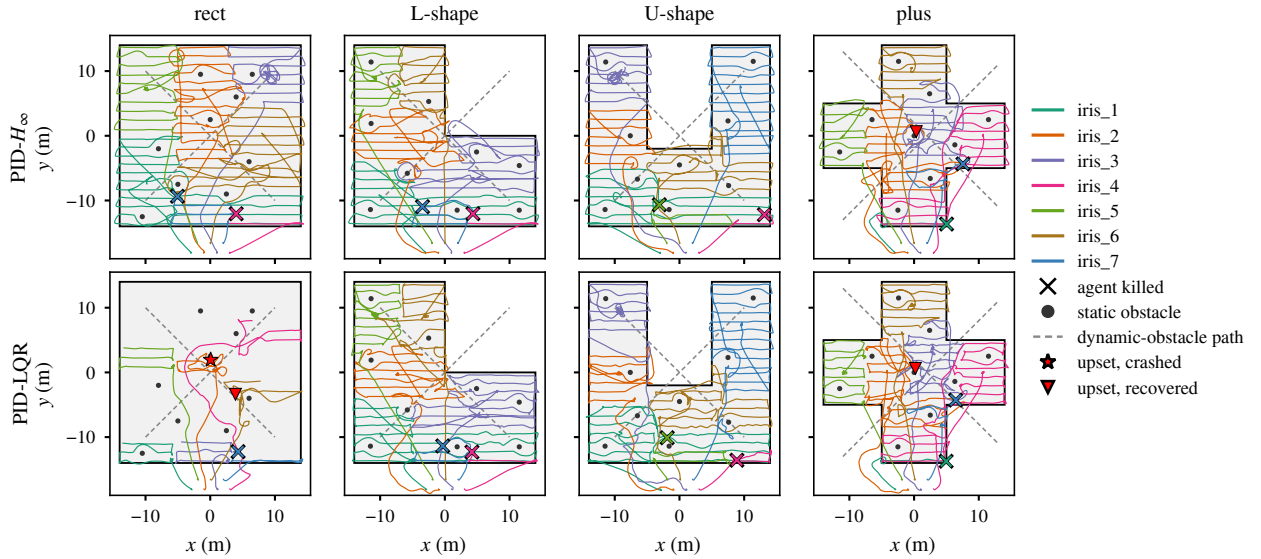


FIGURE 5. Flown trajectories of live agents above 0.5 m, including transit from the launch pad ($y \approx -18$ m). Dark discs: static cylinders; dashed: moving-cylinder paths; crosses: agent kills; red markers: tilt upsets above 30°.

Both controllers use the same planner, filter and plant model; Eq. (2) is evaluated with the LQR outer-loop gains in both cases. Each region-controller pair was flown once ($N = 1$) and recorded on video. The two *plus* missions also include three transit guards added before they were flown (a stall detector, a speed reduction above 16° tilt and a 20 s start-wait timeout). Metrics are computed from 20 Hz controller logs, discarding the rows after an agent is killed and the first 20 s. Lateral error is the deviation from the lane reference during steady east-west sweeps (heading stable for 1.5 s, speed above 0.6 m/s); torque effort is the time average of $\tau_x^2 + \tau_y^2$; obstacle distance is measured from the drone centre to the cylinder surface. The collision geometry of the airframe reaches 0.235 m (body) to 0.348 m (rotor tips) from its centre, so a surface distance below these values means certain or possible contact.

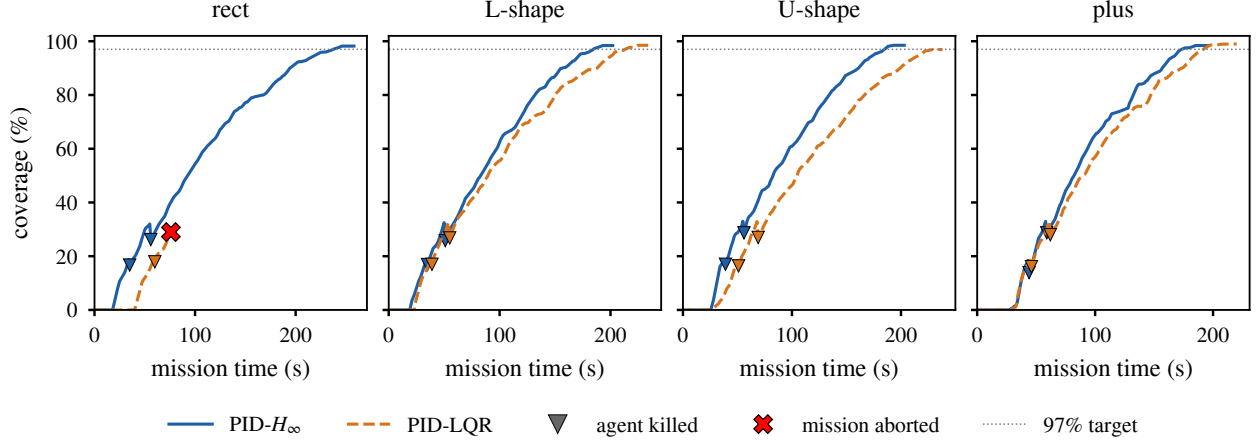


FIGURE 6. Coverage against mission time (coordinator clock, 1 Hz). Triangles mark agent kills; the step after each kill is the erased coverage of the failed cell being re-swept.

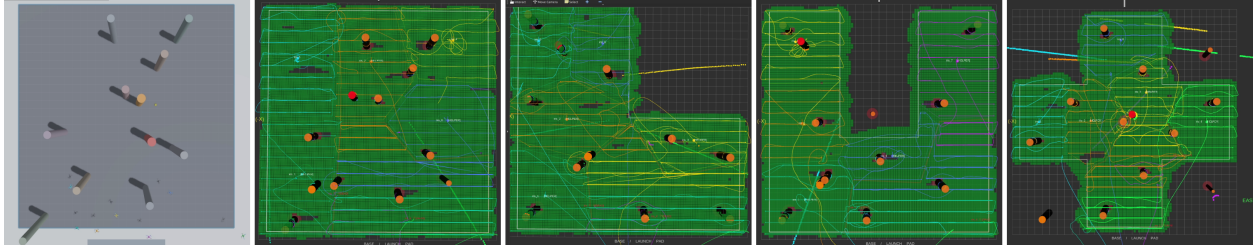


FIGURE 7. (a) Gazebo view of the rect scenario (screenshot); (b)–(e) RViz coverage maps (green) at the end of the PID- H_∞ missions in rect, L-shape, U-shape and plus, with lanes, flown paths and mapped obstacles, extracted from the mission recordings.

RESULTS AND DISCUSSION

Mission Completion and Fault Recovery

Table 2 and Fig. 5 summarise the eight missions. Seven finished with every surviving agent back at its cell centroid and final coverage of 96.9–99.0%, although two of seven agents (28.6%) were lost at 18% and 32% coverage. In every mission the obstacle map contained all nine cylinders (centre and radius within the 0.1 m log resolution) plus at most two unmatched tracks, and no completed mission needed a tier-3 fallback. After each kill the coverage curve drops slightly, because the coverage of the failed cell is erased, and then resumes its slope (Fig. 6); Fig. 7 shows the final maps. The PID-LQR U-shape mission ended just below the 97% target (96.9%) with thin uncovered strips along cell boundaries in the left arm that are nearly absent in the PID- H_∞ map; we did not isolate their cause. The PID-LQR rect mission was aborted at 78.8 s (see below).

Tracking under Turbulence: LQR versus H_∞ Gains

The two designs differ clearly in how tightly they hold a lane (Fig. 8a): the median lateral error was 2.7–3.8 cm with PID- H_∞ and 8.5–8.8 cm with PID-LQR, and the RMS error fell from 11.4–12.2 cm to 7.2–9.7 cm, a 16–39% reduction in all four regions. Most of the difference is a mean offset: PID-LQR flew –5.7 to –7.0 cm downwind (the mean wind blows towards –y), PID- H_∞ within 0.7–1.6 cm of the lane, while the standard deviations differed less (7.0–9.6 vs 9.4–10.8 cm). This is a low-frequency effect: the H_∞ outer integral gain is 3.9 times larger, and the PID-LQR node freezes its position integrator when the error exceeds 0.25 m, which happened in 36–45% of its sweep samples. The mean sweep speed was 1.15–1.21 m/s against 1.02–1.12 m/s for a 1.6 m/s command, which accounts for the 11–15% shorter missions; the 95th-percentile tilt was essentially equal, and so was altitude error outside the two missions with upsets. The linear analysis (Table 1) agrees in direction: the H_∞ gains lower the worst-case gain from disturbance acceleration to position by a factor of 4.0 and the position deviation under the Gauss–Markov input by 3.7 (Fig. 8c).

The price is a higher effort: PID- H_∞ used 3.9–5.6 times the torque effort of PID-LQR in the completed pairs (Fig. 8b), with motor saturation below 0.2% of samples in every mission, and a higher outer-loop crossover (9.08 vs

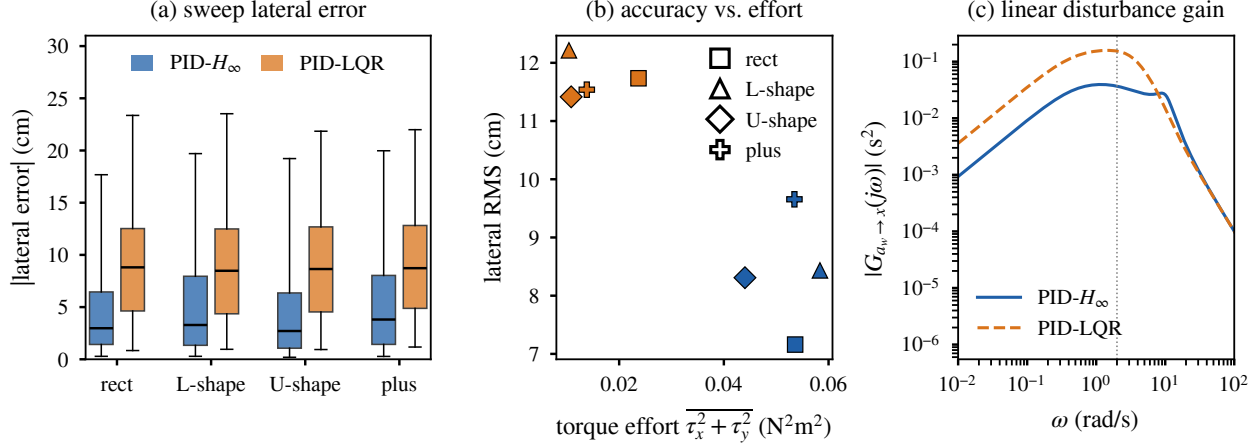


FIGURE 8. (a) Absolute lateral error during steady east–west sweeps (box: quartiles, whiskers: 5th–95th percentiles). (b) RMS lateral error against torque effort per mission. (c) Gain from a disturbance acceleration to position of the linearised outer loop; dotted: corner frequency of the wind model.

4.09 rad/s) with smaller margins. The measured reduction is far smaller than the linear factor because the tilt limit, the reference pre-filter, the QP rate limits and the wind force model are absent from the linear model. As the weights Q , R differ and the integrator gating belongs to the PID-LQR node rather than to the synthesis, the comparison is between two complete gain designs; the contribution of γ is not isolated from that of higher loop gain.

Safety: Clearance, Contact and Upset Events

No two agents came closer than 0.79 m (PID-LQR, L-shape), above the 0.70 m hard radius. The minimum surface distance to static cylinders under PID- H_∞ was 0.49, 0.57 and 0.63 m in rect, L- and U-shape, clear of the airframe, and under PID-LQR 0.06, 0.29 and 0.42 m; in *plus* both controllers passed within 0.24 m of the central cylinder, inside the rotor envelope. The resulting upsets (Fig. 9) share a pattern in the coordinator log: each happened while the agent stepped to its next lane beside a cylinder, after the QP had run in tier 2 for 1–4 s with that cylinder as the binding row. In the PID-LQR rect mission agent 2 came within 0.06 m of the cylinder at (0, 2.5), i.e. body contact, tilted to 155.9° and fell. In *plus*, agent 2 of both missions grazed the central cylinder: under PID-LQR the tilt reached 40.1° with little height loss, under PID- H_∞ 82.5° with a drop to 0.64 m before cruise altitude was regained within about 3 s. With one run per case these events cannot be attributed to the controller; they are better explained by obstacle-blind lanes whose ends and steps fall inside the margin of a cylinder, combined with a barrier built on a 0.22 m disc and a fixed 0.5 m/s^2 wind allowance that does not scale with the measured gusts (peaks of 12.0 m/s). A further 50.8° upset (PID-LQR, rect, agent 6 at 24.4 s) occurred before sweeping began, with the nearest cylinder 0.55 m away, and was recovered. In all 8 missions some tier-2 solves bound by a moving-obstacle row also reported a negative minimum barrier value (down to -0.68 m), but the log keeps only the minimum over all rows and the moving cylinders were not logged, so their physical distance cannot be recovered. A rotor-tip collision envelope, obstacle-aware lane ends, a gust-dependent acceleration budget and logging of the moving obstacles are the next steps.

Limitations

The results rest on one run per configuration with a fixed wind seed, perfect state feedback, perfectly tracked moving obstacles, instantaneous failure detection, a geometric coverage model without camera error, and centralised execution of the upper layers; no component was ablated or compared with a re-implemented baseline.

CONCLUSION

A three-layer architecture combining polygon-clipped Voronoi partitioning, boundary-following sweeps, lane-level fault reallocation, a LiDAR-mapped actuator-consistent CBF-QP and Riccati-synthesised PID control was evaluated in eight simulated missions with seven quadrotors under strong turbulence, unmapped and moving obstacles, and the loss of two agents. Seven missions finished with 96.9–99.0% coverage. Gains from the H_∞ Riccati equation held the lanes 16–39% more tightly and, where both finished, shortened missions by 11–15% compared with LQR gains, at several times the torque effort and with smaller margins; both designs grazed static obstacles under turbulence, and

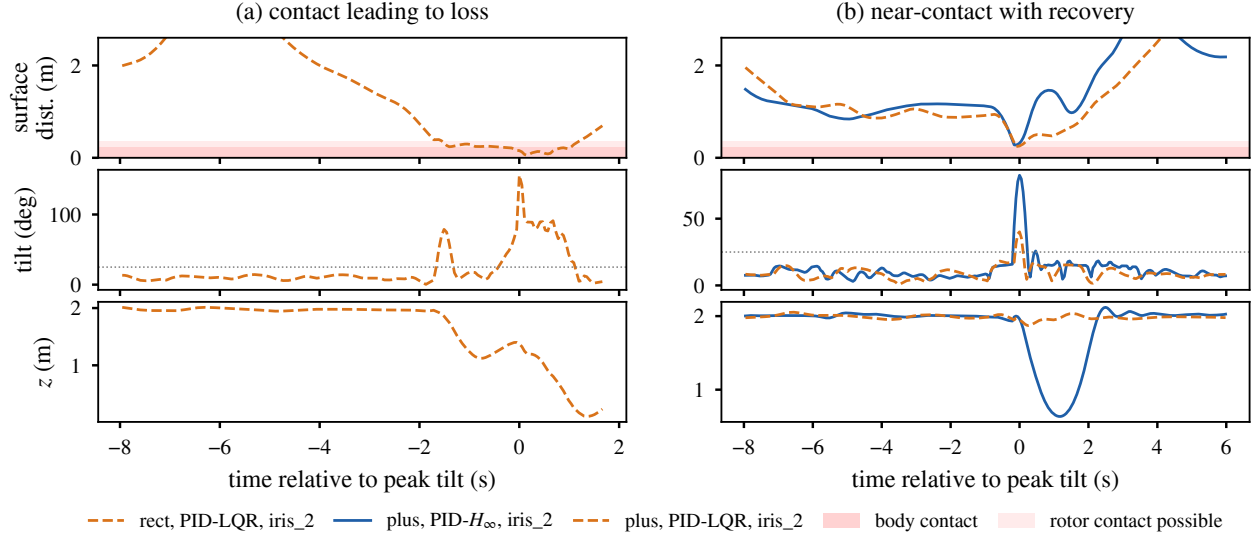


FIGURE 9. Upset events, aligned at peak tilt. Top: centre-to-surface distance to the nearest static cylinder (dark band: body contact, light band: rotor contact possible); middle: tilt (dotted: 25° level-out threshold); bottom: altitude. (a) PID-LQR, rect, agent 2: contact and loss. (b) *plus*, agent 2 under both controllers: grazing contact and recovery.

one contact ended an LQR mission. Future work will repeat the campaign over many wind seeds, enlarge the collision envelope and add wind to the barrier model, distribute the upper layers onto the vehicles, and move to flight tests with on-board state estimation.

REFERENCES

1. S. P. Lloyd, IEEE Trans. Inf. Theory **28**, 129–137 (1982).
2. Y. Bai, Y. Wang, M. Svinin, E. Magid, and R. Sun, IEEE Control Syst. Lett. **6**, 944–949 (2022).
3. Y. Bai, Y. Wang, X. Xiong, J. Song, and M. M. Svinin, IEEE Control Syst. Lett. **7**, 3217–3222 (2023).
4. K. Cao, Y. Chen, S. Gao, H. Dang, and D. An, Appl. Sci. **13**, 8530 (2023).
5. K. Lee and K. Lee, IEEE Access **12**, 5503–5516 (2024).
6. C. S. Tan, R. Mohd-Mokhtar, and M. R. Arshad, IEEE Access **9**, 119310–119342 (2021).
7. S. Hu, T. Xu, and B. Wang, Sensors **21**, 2221 (2021).
8. G. Fevgas, T. Lagkas, V. Argyriou, and P. Sarigiannidis, Sensors **22**, 1235 (2022).
9. M. A. Luna, M. Molina, R. Da-Silva-Gomez, J. Melero-Deza, P. Arias-Perez, and P. Campoy, J. Field Robot. **41**, 1480–1497 (2024).
10. A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, IEEE Trans. Autom. Control **62**, 3861–3876 (2017).
11. K.-C. Hsu, H. Hu, and J. F. Fisac, Annu. Rev. Control Robot. Auton. Syst. **7**, 47–72 (2024).
12. W. Xiao and C. Belta, IEEE Trans. Autom. Control **67**, 3655–3662 (2022).
13. K. Garg, S. Zhang, O. So, C. Dawson, and C. Fan, Annu. Rev. Control **57**, 100948 (2024).
14. M. Tayal, R. Singh, J. Keshavan, and S. Kolathaya, “Control barrier functions in dynamic UAVs for kinematic obstacle avoidance: A collision cone approach,” in *2024 American Control Conference (ACC)* (Toronto, Canada, 2024) pp. 3722–3727.
15. P. Abichandani, D. Lobo, M. Muralidharan, N. Runk, W. McIntyre, D. Bucci, and H. Benson, Drones **7**, 58 (2023).
16. J. C. Doyle, K. Glover, P. P. Khargonekar, and B. A. Francis, IEEE Trans. Autom. Control **34**, 831–847 (1989).
17. Z. Zyadat, N. Horri, M. Innocente, and T. Statheros, Sensors **23**, 1954 (2023).
18. J. O’Keeffe, IEEE Robot. Autom. Lett. **10**, 8954–8961 (2025).
19. I. A. Herdian, E. Ekawati, F. Mukhlis, and P. Prabaswara, J. King Saud Univ. Eng. Sci. **38**, 41 (2026).
20. S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, Sci. Robot. **7**, eabm6074 (2022).
21. N. Koenig and A. Howard, “Design and use paradigms for Gazebo, an open-source multi-robot simulator,” in *2004 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Vol. 3 (Sendai, Japan, 2004) pp. 2149–2154.
22. A. Miranda-Moya, H. Castañeda, and H. Wang, Drones **7**, 700 (2023).